



University of Asia Pacific

Department of Computer Science and Engineering

CSE 316: Microprocessors and Microcontrollers Lab

LAB REPORT

Experiment Number:05

Experiment Title: Water Level Based Automatic Pump Control

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1. Experiment Name : Water Level Based Automatic Pump Control

2. Objective:

To design and implement a water-level based automatic pump control system using Arduino. The system monitors the water level through an ultrasonic sensor. Based on the measured level, the pump is automatically controlled (turned on or off) to maintain the water within a defined safe range.

3. Hardware & Software Requirements:

Hardware:

- 1 Arduino Uno R3
- 1 Ultrasonic Distance Sensor (HC-SR04)
- 3 LEDs (Red, Yellow, Green)
- 1 L293D H Bridge Motor Driver
- 1 9V Power Source
- 1 DC motor
- 1 Breadboard
- 3 Resistors
- 1 USB Cable with power supply
- Jumper Wires

Software:

- Arduino IDE
- Tinkercad

4. Circuit Diagram:

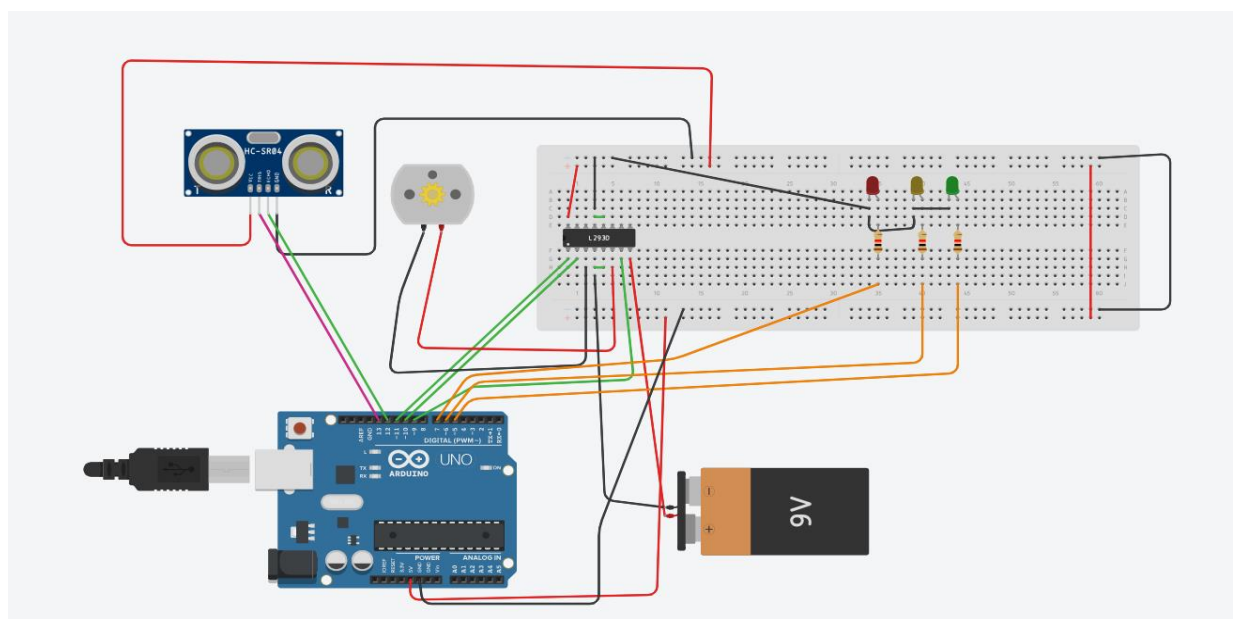


Figure 1: Circuit diagram of a Water Level based Automatic Pump Control

5. Assembly Program:

```
// Water Level Based Automatic Pump Control
const int trig = 13;
const int echo = 12;
const int en1 = 11;
const int inp1 = 10;
const int inp2 = 9;
const int LEDred = 7;
const int LEDyellow = 6;
const int LEDgreen = 5;

void setup(){
  pinMode(trig,OUTPUT);
  pinMode(echo,INPUT);
  pinMode(en1,OUTPUT);
  pinMode(inp1,OUTPUT);
  pinMode(inp2,OUTPUT);
  pinMode(LEDred,OUTPUT);
  pinMode(LEDyellow,OUTPUT);
  pinMode(LEDgreen,OUTPUT);

  Serial.begin(9600);

  delay(300);
}
void loop(){
  //Sending ultrasonic pulse
  digitalWrite(trig, LOW);
  delayMicroseconds(2);
  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig, LOW);

  //Measuring duration of echo
  int duration = pulseIn(echo, HIGH);

  //Calculating distance in cm
  int distance = (duration * 0.034) / 2;
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");
```

```

//Controlling LEDs and Motor based on distance
  if (distance > 150){
    digitalWrite(en1, HIGH);    //motor ON
    digitalWrite(inp1, HIGH);
    digitalWrite(inp2, LOW);
    digitalWrite(LEDred, HIGH);
    digitalWrite(LEDyellow, LOW);
    digitalWrite(LEDgreen, LOW);

    Serial.println("Alert: Low Water Level Reached, Pump ON");
  }
  else if (distance >50 && distance<=150){
    digitalWrite(en1, HIGH);    //motor ON
    digitalWrite(inp1, HIGH);
    digitalWrite(inp2, LOW);
    digitalWrite(LEDred, LOW);
    digitalWrite(LEDyellow,HIGH);
    digitalWrite(LEDgreen, LOW);

    Serial.println("Warning: Reaching Low Water Level");
  }
  else {
    digitalWrite(en1, LOW);      //motor OFF
    digitalWrite(inp1, LOW);
    digitalWrite(inp2, LOW);
    digitalWrite(LEDred, LOW);
    digitalWrite(LEDyellow,LOW);
    digitalWrite(LEDgreen,HIGH);

    Serial.println("Saturated
Water Level, Pump OFF");
  }
  delay(1000);
}

```

6. Observations:

To facilitate observations, LEDs (red, yellow, green) were added to the design. Upon execution in Tinkercad, the ultrasonic sensor continuously measured the water level distance, and the system responded with corresponding LED indicators and pump actions:

a) Distance > 150 cm (Low water level):

- Red LED turns on to signal a near empty tank
- Pump activated automatically via the motor driver to refill the tank.

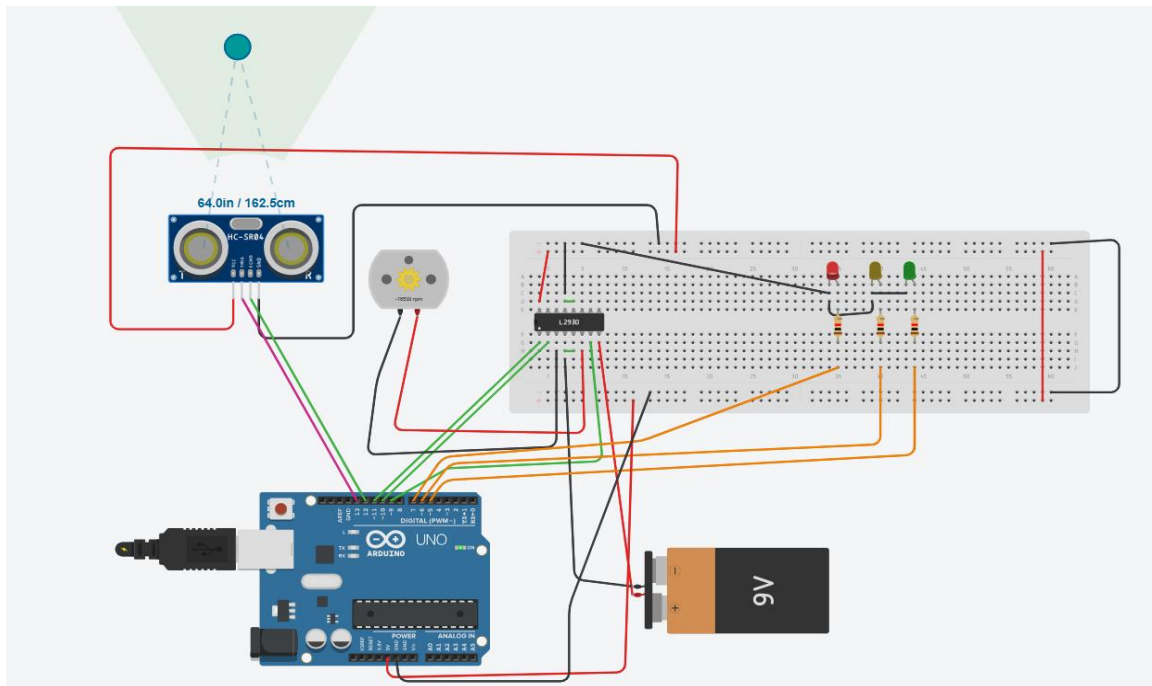


Figure 2: Case A – Full tank, red LED ON

- ### b) $50\text{ cm} < \text{Distance} \leq 150\text{ cm}$ (Medium water level):
- Yellow LED turns on to indicate a warning - Pump remains ON to continue refilling.

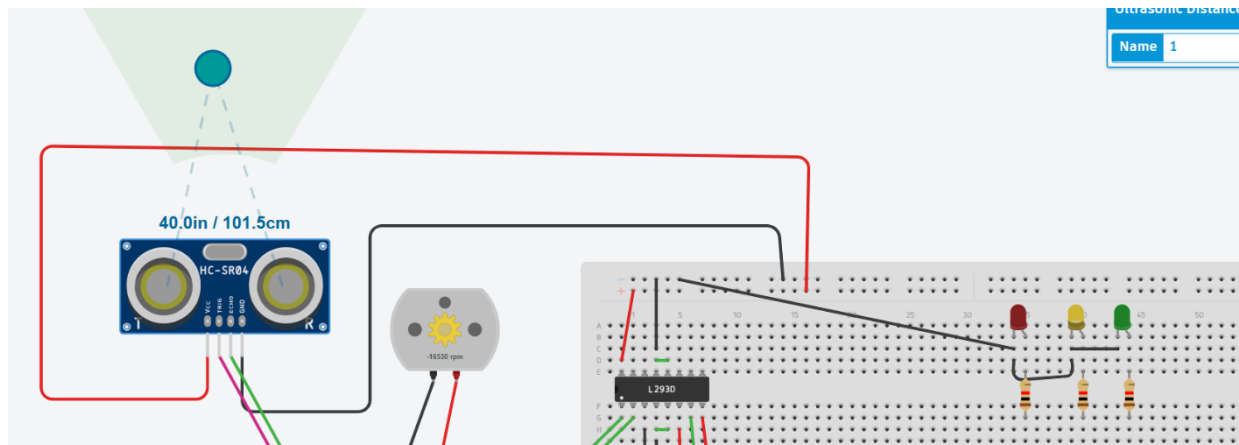


Figure 3: Case B – Nearly full tank, yellow LED ON

c) Distance ≤ 50 cm (Tank full):

- Green LED turns on to signal a full tank
- Pump remains OFF to prevent overflow

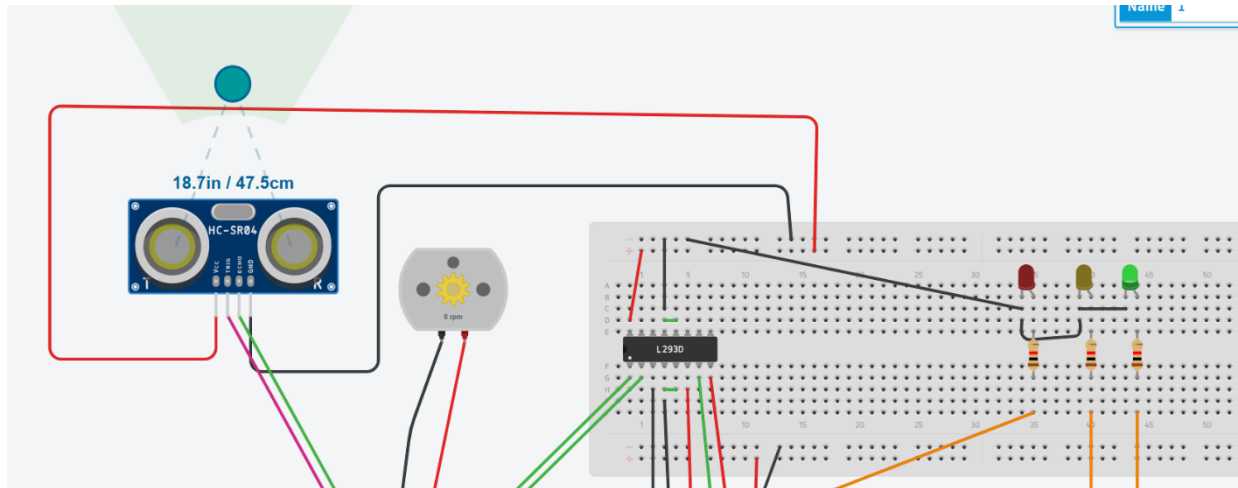


Figure 4: Case C – Empty tank, green LED ON

7. Result:

An automated solution for managing water levels and pump operation was developed using an ultrasonic sensor, LED indicators, and a DC motor controlled through an H-bridge driver. The values observed in the Serial Monitor aligned accurately with the LED signals and the pump's behavior, indicating that the system reacted appropriately to varying water levels. Changes between different operating states occurred smoothly and reliably, confirming the correctness of both the circuit design and the implemented code logic. In summary, the project fulfilled its intended purpose and offered practical experience in developing an integrated automation system combining hardware components with software control for real-life use.

8. Conclusion:

This work demonstrates the capability of microcontrollers to combine data acquisition, logical processing, and system control into a reliable automation solution. Although the prototype was developed on a limited scale, the same concepts can be applied to larger systems such as household water tanks, irrigation networks, and industrial storage facilities. There is ample scope for future enhancements, including the use of variable-speed pumps, improved sensor accuracy through calibration, and the integration of IoT features for remote supervision and alert notifications. In conclusion, the project successfully achieved its objectives and provided valuable hands-on experience in designing practical, real-world automation systems.